

Research Statement

Research Area: Artificial Intelligence, Machine Learning, Robotics, Computer Vision, Image Processing, Bioinformatics, Natural Language Processing, Big Data, Statistical Visualization, Software Engineering, Computational Cognitive Science, Optimization, Quality Control, Reliability, Mechatronics, Industria 4.0.

Research Accomplishment: Highlights of my research accomplishments are : (i) I published thirty five (35) peer-reviewed research articles till date (October 2018). Among them, eighteen (18) papers are published on Computer Vision and Image Processing, five papers (5) on automatizing manufacturing operations by industrial robots using artificial intelligence algorithms, eight papers on welding sequence optimization, three papers on computational cognitive science, and one paper on theoretical machine learning; (ii) My research articles are published in the reputed journals like Pattern Recognition (Impact Factor (I. F.) 3.453), IEEE Signal Processing Letters (I. F. 2.528), American Journal of Neuroradiology (I. F. 3.550), and premier conferences like MICCAI (Acceptance Rate (A. R.) ~32%, rank 1 conference in computer science) , ECML (A. R. ~30%) , ACCV (A. R. ~29%), and ICIP (A. R. ~45%); (iii) I received more than 350 google scholar citations and 3000 research gate reads till date; and (iv) Recently I received a very competitive basic science research grant as young investigator and MITACS industrial internship awards for three times during my graduate studies.

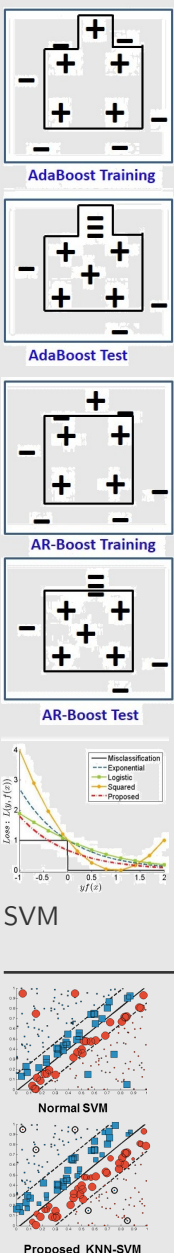
Research Contributions: My research interest lies at the intersection of advancement and utilization of algorithms at various applications that can do both. My interest includes : (i) developing computationally viable solutions to large scale (complex) data; (ii) performing completely automated data analysis; (iii) offering solutions to vision based applications that involve multidisciplinary research (Artificial Intelligence, Machine Learning, Computational Intelligence, Computer Vision, Medical Imaging, Bioinformatics, Statistics, Reliability, Computational Cognitive Science and Software Engineering); (iv) automatizing and optimizing multifaceted cutting-edge industrial applications such as welding, industrial robots, manufacturing, transportation, and oil sand mining. My diverse educational background (Bachelor of Mechanical Engineering, Master in Quality, Reliability and Operations Research (QROR) as well as Master and PhD in Computer Science) facilitates me to conduct research in a broad range of interdisciplinary areas. Highlights of my research contributions are mentioned below.

Machine Learning, Soft Computing, and Big Data

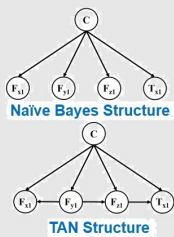
A plethora of emerging Big Data applications such as, social media analysis, video surveillance, network security monitoring, personalized and precision medicine, genomics, neuroscience, economics, finance, and such others, usher processing and analyzing streams of data to extract valuable information in real-time. Big Data hold great promises for discovering subtle population patterns and heterogeneities that are not possible with small-scale data. On the other hand, the massive sample size and high dimensionality of Big Data introduce unique computational and statistical challenges: (i) high dimensionality brings noise accumulation, spurious correlations and incidental homogeneity; (ii) high dimensionality combined with large sample size creates issues such as heavy computational cost and algorithmic instability; (iii) the massive samples in Big Data are typically aggregated from multiple sources at different time points using different technologies. This creates issues of heterogeneity, experimental variations and statistical biases, and requires us to develop more adaptive and robust procedures. These challenges are distinguished and require new computational and statistical paradigm. In terms of statistical accuracy, dimension reduction and variable selection play pivotal roles in analyzing high-dimensional data. This motivates advancement of new regularization methods and sure independence screening. In terms of computational efficiency, Big Data motivate the development of new computational infrastructure and data-storage methods. Such a paradigm change has led to significant progresses on developments of fast algorithms that are scalable to massive data with high dimensionality. My research interests lie in capacitating classification algorithms for handling Big data by incorporating data-driven regularization strategy which improves classification accuracy by avoiding overfitting and expediting the convergence. Details are given below.

Adaptive Regularized Boosting (AR-Boost): Adaboost is one of the state-of-the-art classifiers for two class classification. I developed a novel Adaptive Regularized Boosting (AR-Boost) by incorporating a regularization parameter into the boosting loss function that relaxes the *weak learner assumption* of boosting algorithm and thus facilitates a natural extension of boosting algorithm for multiclass setting [1]. I advocated a regularized exponential loss function for AR-Boost which enforces more weight than Adaboost on misclassified samples at each iteration to classify correctly in the immediate iteration and consequently leads to early convergence. Unlike Adaboost, the user can select optimal weights of the features through cross-validation. Currently, I am deploying AR-Boost for pediatric brain age classification from MRI which can help in early non-invasive diagnosis and prognosis of abnormal pediatric brain development.

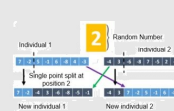
kNN Regularized Support Vector Machine (kNN-SVM): We describe a novel approach to combining the properties of two well-known formalisms: k-Nearest Neighbors (kNN) and Support Vector Machines (SVMs). We introduce a k-Nearest Neighbor (kNN) based regularization term into SVM optimization framework which improves the classification accuracy of SVM by leveraging the strengths of both SVM and kNN. The proposed loss function minimizes the slackness and maximizes the surplus, provides weights to the slack variables based on the degree of their k-NN and thus introduces a novel regularization strategy into SVM framework. Our approach extends the total margin SVM, which considers the distance of



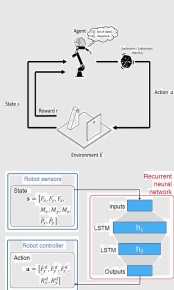
BN



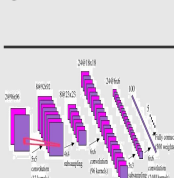
GA



RL

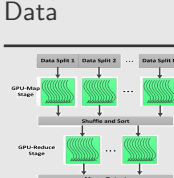


CNN

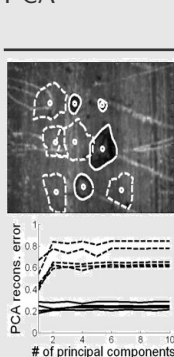


Big-

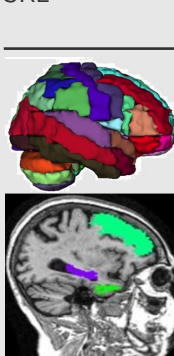
Data



PCA



SRL



all points from the margin; each point is weighted based on its k neighbors. The intuition is that examples that are mostly surrounded by similar neighbors, i.e., of their own class, are given more priority to minimize the influence of drastic outliers and improve generalization and robustness. Thus, our approach combines the local sensitivity of k NN with the global stability of the total-margin SVM. We analyzed its theoretical properties and empirically demonstrate improved generalization and robustness in noisy settings for both binary- and multi-class classification. We are now extending the idea of k NN-SVM in Big Data domain by ameliorating the bottlenecks of k NN in high dimensional framework.

Bayesian Network (BN) Classifier: Motivated by the success of Bayesian classifier with with strong assumptions of independence among features, called Naive Bayes (NB), we deployed Bayesian Network (BN) classifier with slightly less restrictive assumptions of independence among features, called Tree Augmented Naive Bayes (TAN) to classify successfully the robot execution failures associated with automation of peg in hole insertion task by Industria Robot, Motoman MH-6 [2]. TAN approximates the interactions between attributes by using a tree structure imposed on the NB structure that generalize the NB classifier and explicitly represent statements about independence which outperforms Naive Bayes, yet at the same time maintains the computational simplicity (no search involved) and robustness that characterize NB.

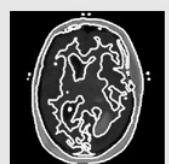
Exploiting Domain Knowledge for Early Convergence of Genetic Algorithm (GA): Genetic algorithm (GA) is a heuristic search and optimisation technique inspired by natural evolution. They have been successfully applied to a wide range of real-world problems of significant complexity. Success of GA depends on smart choice of selection, crossover and mutation mechanism and fitness function as well. We advocated domain knowledge in selection, crossover and mutation algorithms that expedites the convergence and makes GA more robust for combinatorial optimization problems. We developed a novel restricted weak integer composition based sample initialization algorithm which introduce diversity in the population and in turn increases likelihood of convergence for solving progressive censoring through GA. In another effort, Algebraic Method (AM) is a well known analytical method for robot inverse kinematic problem. However, this method demonstrates poor convergence rate and falls into local minima in many occasions. We developed a GA based solution where AM has been introduced to reduce the search space and thus helps in speeding up the convergence rate for GA. Like Differential Evolution (DE), we introduced a novel vector differential based crossover algorithm in both binary and continuous domains which increases the likelihood of finding global optima. We also advocated data driven parameter optimization method for GA [3]. We are now performing parallel and online implementation for GA to offer realtime solutions.

Tailoring Classification Algorithm for Big Data: Support Vector Machine (SVM) is currently the state-of-the-art model for many classification problems but they suffer from the complexity of their training algorithm which is at least quadratic with respect to the number of examples. Conventional Sequential Minimization Method (SMO) is not computationally efficient for using Support Vector Machine (SVM) for large scale problems. We are conducting research on facilitating SVM for solving large scale problems using coordinate descent method. We are also introducing the idea of mixture of experts, divide and conquer, Bayesian committee machine, and MapReduce algorithms in classification algorithms such as SVM, AdaBoost, and Decision Tree for enabling them tackle to Big Data where training data is divided into a number of clusters, run training algorithm on each of the cluster separately and parallelly and then amalgamate them to generate the training model using the mixture of experts. In an alternative procedure, we are implementing a novel clustering based sampling techniques to generate a representative sample of the Big data and perform training algorithm on the sample instead of the whole large dataset to alleviate massive computation.

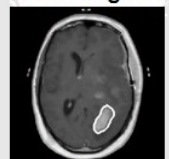
To Improve Medical Diagnosis through Decision Tree: In medical decision making (classification, diagnosing, etc.) there are many situations where decision must be made effectively and reliably. Conceptual simple decision making models with the possibility of automatic learning are the most appropriate for performing such tasks. Decision trees are a reliable and effective decision making technique that provide high classification accuracy with a simple representation of gathered knowledge and they have been used in different areas of medical decision making. In addition, automated decision making models developed for medical applications need to be interpretable akin to decision tree that help medical practitioners for making diagnostic and prognostic decisions for treating diseases. We are deploying decision tree for making diagnostic decisions for Non-Alcoholic Fatty Liver Disease (NAFLD). Automated decision tree can successfully diagnose NAFLD in 96% cases using only biochemical features. However, the conventional clinically-oriented metabolic feature based decision tree developed by exploiting the knowledge of the physician (domain expert) can classify accurately only in 70% cases. We implemented lookahead decision tree that can compensate immediate wrong decision as well as incorporate induction of XOR along with conjunction and disjunction.

Improving exploration in Reinforcement Learning (RL) through Domain Knowledge: We presented a novel work on how to improve exploration in reinforcement learning using domain knowledge and knowledge-based approaches to reinforcement learning. Reinforcement learning algorithms normally face the problem of deciding whether to execute explorative or exploitative actions, and the paramount goal is to limit the number of executions of suboptimal explorative actions. We achieve this through better understanding of domain knowledge and understanding of algorithms' and domains' properties. We incorporated the knowledge of thermal affected zone and joint rigidity of the system while solving welding sequence optimization through RL [4]. At present, we are deploying monte carlo tree search for welding sequence optimization where exploration-exploitation dilemma is being balanced by UCB (Upper Confidence Bound).

Deep Reinforcement Learning (DRL): Motivated by unprecedented success of human level control through DRL [5], we

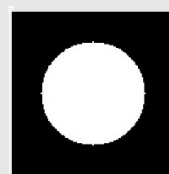


Without bounding box

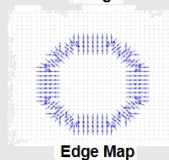


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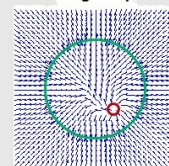
Snake



Image

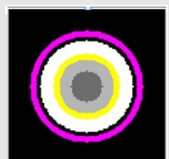


Edge Map

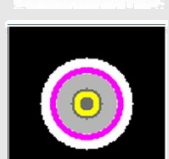


Diffusion with Dirichlet boundary condition

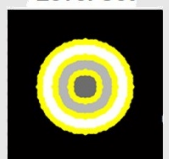
HLS



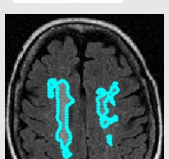
Initialization



Level Set



Hierarchical Level Set



are conducting research on developing a novel object detection algorithm through DRL. The agent (proposed DRL based object detection algorithm) automatically learns a policy which computes the probability of taking an action (translation and scale) in every state so that the agent can reach to the final state (desired object circumscribed by the bounding box) from the initial state (the whole image itself is circumscribed by the bounding box). DRL is adept in reducing run time complexity without compromising the detection accuracy by avoiding searches in the areas where the probability of having objects is very low. The agent constructs a deep learning feature representation of the current state (bounding box), which is then used by the Deep Q-Network (DQN) to determine the next suitable course of action so that the goal can be achieved very quickly. We are also working for developing DRL based model for robot inverse kinematic problem and peg in hole automation. Reduction of state space is one of main challenges in these problems which are being handled through DRL.

Computer Vision, and Image Processing

I developed several novel image segmentation, and tracking algorithms. Very few of them are mentioned below.

Deep Convolutional Neural Network (D-CNN): In recent years, D-CNNs have achieved great success in many problems of machine learning and computer vision due to plethora of computational power and data availability. We have implemented the idea of user interactive segmentation by random walk on a complete graph where feature vectors are learned through convnet. We developed and implemented a novel D-CNN based completely automated image segmentation algorithm where the novelty lies in two fronts: doing random walk on a complete graph and learning feature for similarity measurement by convnet. Proposed algorithm obtains state-of-the-art performance.

Statistical Relational Learning (SRL): We developed a SRL based classifier for three class classification of Alzheimer disease [6]. SRL is a natural choice of model relations, it can find the relationship between the different regions of the brain representing the classes Alzheimer Disease (AD), Cognitively Normal (CN), and Mild Cognitive Impairment (MCI). Our extensive experiments demonstrate that Statistical Relational Learning (SRL) outperforms propositional classifier such as Support Vector Machine (SVM), Decision Tree (DT) for AD classification. In future, SRL can be used for modeling, analyzing and simulating complex brain networks helping diagnosis and treatment plan for a lot of diseases and thus Artificial intelligence initiates a new horizon in personalized medicine.

Change Detection or Fast Bounding Box (FBB) Method: We developed a novel region-based and histogram-of-features based technique for change detection. Encouraging comparative results with the change detection or background subtraction algorithms can be attributed to a region-based approach as opposed to pixel-based change detection. Histogram is forgetful about the spatial location of the features. Our method takes advantage of this feature and is able to correctly identify the change even when there is considerable shift between objects in the test and the reference images. Interestingly enough, we show that our energy functional formulation for change detection allows variational optimization via level set-based curve evolution for both unsupervised and supervised binary image labeling. This aspect certainly demands further exploration of the proposed energy functional. The optimization of the proposed energy functional simply consists of two very efficient searches if a crude segmentation such as a bounding box around the region of change is sufficient for many applications akin to Brain Tumor and Edema Detection.

Currently, many clinical centers maintain large archival databases of Magnetic Resonance (MR) images of brain tumors for various purposes, as this information may help physicians and radiologists to diagnose and treat new patients - e. g., by determining the effectiveness of various treatments on previous patients with similar tumor or edema volumes. As these archived MR image databases may contain a large number of studies, it is important to index their contents effectively. If these tumors were segmented, the databases could be indexed based on various aspects of the tumor, such as its size, location and type. However, accurate tumor boundary delineation is a very complex task to be accomplished accurately. We have developed an automatic, fast, and approximate segmentation technique that deals with the problem associated with accurate tumor segmentation, by locating a "bounding box" - i.e., axis-parallel rectangles, around the tumor or edema on a magnetic resonance slice exploiting the left-right symmetry of human brain [7]. We can then use this bounding box to determine tumor position, type and size to retrieve historical cases relevant to the current patient and help formulate treatment plans using deep learning. The resulting method is very fast, robust and reliable for indexing tumor or edema images for both archival and retrieval purposes and it can be used as a vehicle for further clinical investigations. Our papers received more than one hundred and fifty google scholar citations till date (October 2018).

Automation of Active Contour or Snake Algorithm: Snake is revered as state-of-the-art interactive segmentation algorithm for microscopic cell images. The introductory paper on snake has secured more than nineteen thousand google scholar citations till date (October 2018) [8]. However, snake fails to delineate object boundaries in many real-life applications which urge automation. Literature demonstrates that snake automation needs two sequential steps: initialization and evolution. However we ameliorated snake automation by integrating a self-validation mechanism into snake segmentation framework to revolutionize snake from interactive to automatic segmentation. We fostered three novel algorithms towards snake segmentation: (a) A quadtree or Fast Bounding Box (FBB) based fast object localization method to initialize snake; (b) Interleave Directional Gradient Vector Flow (IDGVF) where novelty lies in two fronts: (i) instills Dirichlet Boundary Condition (DBC) into existing snake evolution algorithm which transmutes the gradient field anisotropic and snake is manifested as more initialization independent; (ii) computation of gradient or force field and snake evolution algorithm is performed concurrently in an interleave fashion which fortifies snake over noises; (iii) evolved snake is sieved through a classifier such as Principal

My future endeavor involves extending such techniques even to objects having low boundary contrast, so that automated active contour algorithms would capacitate delineate object boundaries under poor lighting conditions (which is prevalent in many typical industrial setups) and from low-contrast cellular/molecular imagery. One of my other future works include developing generalized object detection and tracking framework by incorporating self-validation mechanism with spatiotemporal features. One of its significant applications include cell detection and tracking. We are also intending to address few more important questions such as: can we have an unsupervised or semi-supervised classification paradigm for this self-validation mechanism for automatic segmentation? Undoubtedly, the unsupervised/semi-supervised learning can curtail the difficulties of frequent supervised learning in applications where object characteristics change considerably and frequently.

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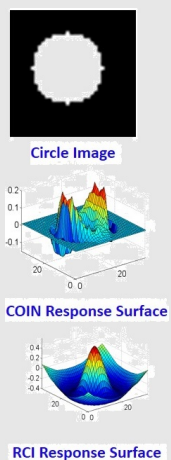
IF



within each group separately. Subsequently, coarse registration is conducted to find the global anchor image selected among the whole time sequenced images and then a deformable registration is conducted on the whole sequence. Two-stage MST-based deformable registration algorithm can incorporate larger drifts and distortions more accurately than conventional one shot registration algorithm by fine-tuning the larger amount of deformation incrementally in a couple of stages. Our method outperforms other methods on both 2D and 3D microscopy image sequences of mouse arteries used in atherosclerosis study.

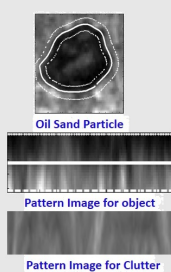
Inclusion Filter (IF): We developed inclusion filter which is used to remove noises and fill the holes from the images before applying the segmentation algorithm [14]. The design of inclusion filters is motivated by self-duality. Self-duality (gray-level inversion invariance) is a desired criterion for many real-life applications, such as oil sand mining where objects appear as both darker and brighter than the background. The inclusion filter is a special class of connected filters which removes minor interior regions and clutter based on the connected component relationships defined by the adjacency forest. Connected filters offer important edge-preserving qualities which is very critical for segmentation. We demonstrated that the use of the inclusion filter significantly improves the edge fidelity and the insensitivity to initialization of active contour or snake algorithm for the oil sand and lung cavity segmentation from magnetic resonance imagery.

RCI



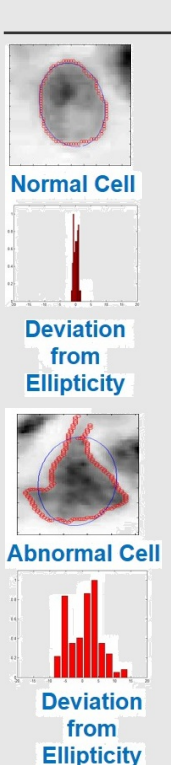
Robust Convergence Index Filter (RCI): CONvergence INdex (COIN) filter, a successful tool for cell localization, evaluates the degree of convergence of the gradient vectors within the neighborhood (region of support) toward a pixel of interest. All previous efforts were to increase the adaptability of the region of support to make the COIN filter robust and accurate. However, improving the quality of the image gradient map was ignored, which results in poor performance of the members of the COIN family in noisy settings. We developed a new Robust Convergence Index (RCI) filter that tailors the COIN filter in a noisy environment by (a) spreading the gradient vectors within non-homogeneous object regions by convolving an Aggregated Edge Probability Map (AEPM) with an edge preserving gradient vector kernel, and (b) increasing the convergence of the gradient vectors through the integration of the sine and cosine distribution as well as the magnitude of the gradient vectors [15]. AEPM is computed through the consensus of the responses of a number of edge detectors over a wide range of scales, which lessens the effects of clutter by enforcing higher weights to the actual edges, and a non-parametric Kernel Density Estimation (KDE) is used to compute the edge probability map. RCI obtains state-of-the-art performance on histopathological breast cancer cell imagery.

PI



Pattern Image (PI): I developed a novel image feature that has been named as "Pattern Image" that could detect objects successfully under noisy settings. "Pattern Image" unfolds the annular band across the object boundaries to form a rectangular pattern for computational convenience that can discriminate successfully objects (leukocyte cells from *in-vivo* microscopic images used for inflammation study and oil sand particles useful for oil sand mining applications) from clutter [16]. This proposed pattern image carries texture information as well as intensity transition (dark to bright transition and vice versa) across regions of object boundaries and it outperforms other edge based features such as Gradient Inverse Coefficient Of Variation (GICOV), Average gradient strength and sundries.

BRF



Biologically Relevant Features (BRF): We devised novel biologically relevant features useful for predicting breast cancer disease survival and recurrence from histopathological images. Towards achieving this goal, we developed three mathematical features such as (a) *Deviation from ellipticity*, (b) *Cell co-location pattern*, and (c) *Mitotic count* that translate three distinct biologically relevant features (i) *cellular pleomorphism*, (ii) *tubule formation* (randomness and abnormal aberrations of the epithelial cell structure), and (iii) *mitosis* that indicate the presence of cancer respectively. Pathologists determine the grade of the cancer of the patients based on these biologically relevant features. Our mathematical features could replicate the grading offered by pathologists that could avoid the inter- and intra-observer variability and predict the breast cancer disease survival and recurrence from histopathological images.

Imaging Genomics in Cancer Research

In recent years, a new direction in cancer research referred to as "radiogenomics or imaging genomics" has emerged to address methodologies of associating imaging features (imaging phenotype or radiophenotype) with genomic data (molecular phenotype or genotype). Recently, I received a basic science research grant for three years to find the potential association between imaging phenotype and genotype for the treatment of breast cancer. Late stage patients clearly benefit from more aggressive treatments such as chemotherapy, while for early stage patients the toxicity of the treatment may predominate the benefit. Early-stage cancers are the target of the Oncotype DX (genomic) test to find distinct subgroups within the early-stage group with different recurrence probabilities that urge for chemotherapy. However, the Oncotype DX test is very expensive but histological imaging is less expensive. We study that both histopathological findings and molecular alterations that reflect the biological properties of individual primary breast carcinomas and they demonstrate strong association among them. Prognostic decisions with only histological imaging for early stage patients could reduce the treatment cost significantly. We advocate Deep CNN to delve the potential associations among imaging and genomic biomarkers.

Machine Learning and Computer Vision in Manufacturing Industry

The nature of manufacturing systems faces ever more complex, dynamic and at times even chaotic behaviors. In order to being able to satisfy the demand for high-quality products in an efficient manner, it is essential to utilize all means available. One area, which saw fast pace developments in terms of not only promising results but also usability, is machine learning. In order to meet the challenges of ever-growing demand for high-quality products in an efficient manner, modern industry practices are being supported by integrating cyber-physical systems, biotechnology and emerging digital technologies such



as remote sensing, cloud computing, cognitive computing, and Internet of Things (IoT), leading to the notion of "Industria 4.0" or "Smart Manufacturing" or "Smart Factory". These emerging digital technologies contribute to better analyzing and understanding complex, multivariate and unpredictable production systems by monitoring and measuring continuously variable aspects of the physical environment. As a result of these, nowadays industry is experiencing an unforeseen plethora of available data comprising a variety of different formats, semantics, quality, e. g. sensor data from the production line, environmental data, machine tool parameters and such others. This implies a pressing need for large-scale collection, storage, pre-processing, modeling and analysis of huge amounts of data, coming from various heterogeneous sources and leading to the notion of "Big Data". The availability of quality-related data offers potential to improve process and product quality sustainably, quality improvement initiatives, manufacturing cost estimation, process optimization e.g., monitoring, control, scheduling, diagnostics, troubleshooting and overall better understanding of the customer's requirements. Highlights of my contributions in manufacturing are mentioned below.

Development of Machine Learning (Data Driven) based Controller

Many industrial tasks performed by robots require position-force control. In practice, engineers have to tune the force control gains on a case by case basis, in order to adapt to the different stiffness, damping or clearance of work pieces (environment). In contrast to the non-trivial gain tuning of conventional controller, since data-driven learning allows to reduce the amount of engineering knowledge, we advocate a different approach and speed up learning by extracting more information from data. In particular, we learn a probabilistic, non-parametric Gaussian process transition model of the system. By explicitly incorporating model uncertainty into long-term planning and controller learning our approach reduces the effect of model errors, a key problem in model-based learning. It turned out that proposed technique leads to a significant improvement over knowledge-based solutions. The reason is that the machine learning is not limited by the imagination of an expert, but is directly evaluating the process behavior instead. In addition, data driven controller outperforms in unforeseen situations.

Combining Sensors to Improve the Accuracy of the Controller

Designing Arduino based controller for controlling the level of the water in the tank using ultrasonic sensor is a well-known problem. I with my students have incorporated multiple sensors: ultrasonic, visual sensor (camera) and temperature sensor to control the level and temperature of the water in the tank which in turn increase the accuracy of the controller. Similarly, line following by robot using infrared sensor is a popular problem. We developed a Raspberry based controller using multiple sensors: infrared, camera and ultrasonic that can deviate from the line in the presence of obstacles and again return and follow the line. We implemented camera calibration algorithm and utilized kalman filter to combine multiple sensors.

Automation of Industrial Robots for Manufacturing Operations

Automatizing Peg in Hole Insertion Task by Industrial Robot: Parts mating, peg in hole insertion or assembly operation is the most common operation in industry production, but autonomous execution by robots can significantly increase overall productivity. Peg in hole insertion task is a topic largely addressed and long standing problem in robotic research, the popularity of which is not only due to its importance in many industrial assembly tasks, but also for its complexity as a control problem that requires both position and force regulation. We developed both computer vision and without computer vision based algorithms to automate the peg in hole insertion task. We implemented a fast template matching algorithm and followed by non-linear camera calibration method to automatically finding the peg and hole in the complex working environment [17]. However, occlusion and imprecise sensor make the automation task difficult in several occasions. Hence, we portrayed the problem of peg in hole insertion search as a travelling salesman based grid search problem and we automatically look for the work pieces through force-torque sensor built in robot end-effector. Then, we developed both impedance and machine learning based controller for regulating force control. We also offered fast and accurate convergence of robot inverse kinematic solutions by incorporating domain knowledge into novel multi-objective genetic and differential evolution algorithms [18]. We devised a second order autoregressive gaussian process for learning the force-torque signal to accurately classify the robot failure execution towards automation of peg in hole insertion task. We also exploited the statistical transformations of time-series data in this context which allows to reduce the dimension and unearth the discriminative features for the classification task and hence increases the predictive accuracy and reduces learning time. We executed the real experiment on Motoman MH-6, gantry or cartesian, and Crustcrawler AX-18 industrial robot. We also facilitated the use of Motoman SDA-20 dual-arm robot equipped with two three fingered gripper and a force/torque sensing capability for human like operations such as the peg-in-hole tasks by learning contact states using ANN Fuzzy ARTMAP architecture.

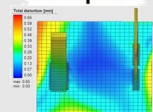
Design a Novel Adaptive and Reconfigurable Robot Gripper using FEM: In another effort, with one of my master students, we developed and implemented a novel Finite Element Method (FEM) based modelling for single degree-of-freedom adaptive and reconfigurable four-fingered robot gripper for automating industrial applications. FEM based deformation and stress analysis facilitates: (a) minimizing the cost of the gripper by selecting cost-effective material that can grab a wide range of objects, and (b) estimating the remaining useful life of the critical components through fatigue analysis.

Probabilistic Visual Slam: We developed a modified Extended Kalman filter and particle filter algorithm as well as we exploited weighted Harris Corner and Sift based point features for visual Simultaneous Localization And Mapping (SLAM) algorithm. We evaluated the performance of our SLAM implementation from two perspectives. First, we compared the robot's estimated trajectory with respect to the ground truth. Second, we evaluated the quality of the generated map by comparing distances between landmarks suggested by the generated map with the corresponding physical distances. Our results illustrate that the estimated distances between landmarks are reasonably close to the groundtruth. Also, the relative

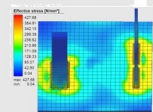




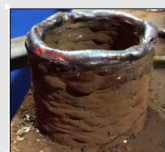
Workpiece



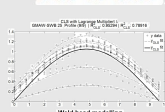
Deformation



Stress



WAAM



Cognitive



NLP

Ram is a good boy. He is reading.

Existing approaches

Ram is a good boy. He is reading.

Our approach

Ontology based Valency Analyzer of Verbs for Anaphora Resolution

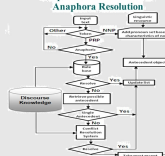
১. তৈরি করেছি একটি ভাষা, যাতে যাঁরা যাঁরা ...

২. take the [power] from the table, I will cut [it] ...

৩. তৈরি করেছি একটি ভাষা, যাতে যাঁরা যাঁরা ...

৪. take the sweet from the [table], I will wash [it] ...

Proposed Architecture for Anaphora Resolution



Evaluation Metrics for Anaphora Resolution System

Metric	System	Score
ACC	System 1	0.85
ACC	System 2	0.82
ACC	System 3	0.80
ACC	System 4	0.78
ACC	System 5	0.75
ACC	System 6	0.72
ACC	System 7	0.70
ACC	System 8	0.68
ACC	System 9	0.65
ACC	System 10	0.62
ACC	System 11	0.60
ACC	System 12	0.58
ACC	System 13	0.55
ACC	System 14	0.52
ACC	System 15	0.50

distance estimated between selected landmarks in the map converges to a stable value which is close the ground truth.

Automation and Optimization of Welding Process

Welding Sequence Optimization: Welding is the most common fabrication process which is extensively used in a wide range of industries such as automotive, shipbuilding, aerospace, pipelines, heavy and earth-moving equipment. Welding deformation and effective (residual) stress derive several negative impact to the manufacturing process and add additional cost in various ways, such as constraints in the design phase, extra operations, cost of quality and overall capital expenditure. With one of my PhD students, we developed novel modified Lowest Cost Graph Search (LCGS) [19], multi-objective Genetic Algorithm (GA) [3] as well as Reinforcement Learning (RL) [4] based Welding Sequence Optimization (WSO) techniques for reducing welding deformation and residual stress. We first demonstrated in literature the effectiveness of WSO using the the combination of conventional and Low Temperature Transformation (LTT) welding wires in the same weldment. LTT increases the fatigue life by introducing compressive residual stresses.

Mathematical Modelling of Weld Bead in WAAM: We developed a novel nonlinear regression followed by Taguchi method for modelling of welding bead geometry in Wire and Arc Additive Manufacturing (WAAM). WAAM has been evolved as an alternative to traditional manufacturing specially for fabricating large aerospace metal components that feature high buy-to-fly ratios. Bead modelling determines the optimum weld settings corresponding to the desired bead geometry. The strength, density and homogeneity of the deposited material depends on the bead geometry. Controlling the bead geometry of the workpiece is much easier and is used to predict its mechanical properties.

Computational Cognitive Science

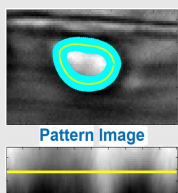
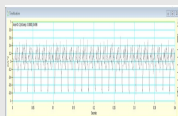
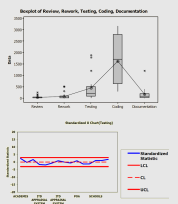
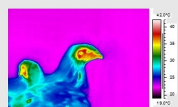
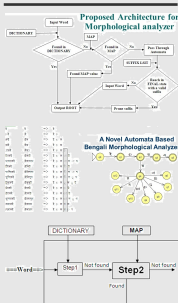
As a statistician it was a matter of curiosity for me to measure mental and psychological processes. Statistical models are being applied to various disciplines outside engineering. Another thematically distinctive area of research is Applied Psychology. I felt motivated to develop statistical models, an also understand how socio-psychological processes in behavior may be predicted. This kind of research also has relevance for artificial intelligence and computational systems of perception and simulation. I participated in experiment design and analysis for three different studies: (i) color perception: we evaluate cognitive perception of the different segments of the population on the color "Rosa Mexicano", which is widely used in art, fashion, painting, architecture, and crafts [20]; (ii) statistical measurement of perception in interactive game: we designed an interactive media to check whether a short real time intervention could act as a mood changer for adolescents, in a stressful learning scenarios such as classroom and respondents' reaction were collected on a combined psychological emotional-valence scale [21]; and (iii) Computer Generated Biological Imagery (CGBI) to stimulate horripilation: we studied the specific effect of visual imagery of physical sensational horror which aims at evoking effects of chills, or goosebumps involving nervous shudder and withdrawal from objects of contemplation. Statistical tools and techniques (factor analysis, ANOVA), visualization (pie-bubble chart), and image processing algorithms have been implemented in these projects.

Anaphora Resolution and Discourse Analysis in Resource-Scarce Languages

In the last decade, there are a lot of researches on Anaphora Resolution (AR) in Indo-European languages can be found in literature. However, it is hardly found any system or sufficient data for AR in resource-scarce languages like Indian languages in public domain. A dedicated conference in this domain "Discourse Anaphora and Anaphora Resolution Colloquium (DAARC)" was initiated to share the resources and knowledge among the community but at present it is not in active. To facilitate developing computational tools and sharing the resources with the community in resource scarce languages like Indian and Spanish languages, at present we have focused our researches into following three subareas of natural language processing.

Corpus Analysis for Natural Language Processing: In this research, we have borrowed the domain knowledge of linguistics and addressed issues like pronouns with origin, etymological history, classifications, various features and exceptions, syntactic and semantic study and have performed a quantitative corpus analysis of Bengali pronoun for natural language processing. The morphological structure of Spanish and Bengali pronouns has also been explored. Different bench mark Spanish and Bengali corpora have been used for this study. This research also presents the frequency of pronouns and the coverage in the corpus. The well known LL test is adopted for the significance test. Finally, the relative rank list of pronouns is computed and the list shows that the personal pronouns dominate in the language.

Anaphora and Coreference Resolution: Anaphora Resolution (AR) which most commonly appears as pronoun resolution is the problem of resolving references to earlier or later items in the discourse. These items are usually noun phrases representing objects in the real world called referents but can also be verb phrases, whole sentences or paragraphs. The research on Anaphora Resolution is more than thirty-five years old in the field of Computational Linguistics. However, the problem is still considered as an uphill battle in natural language processing (NLP) and related areas like Question Answering, and Machine Translation. we intend to develop a knowledge-based classifier for anaphora resolution in resource-scarce languages by incorporating the domain knowledge of linguistics into the machine learning based classifier such as support vector machine, decision tree and AdaBoost. We also performed an empirical study for end-to-end (post-tagger, named entity resolution, chunking) anaphora resolution system in Bengali and conducted analysis of the propagation of uncertainty at each stage of the anaphora resolution system. Then we developed a novel pronoun emitting approach for Bangla anaphora resolution. Unlike the dominant trend of pronominal anaphora resolution (PAR) by looking for the right antecedent after a pronoun is encountered, we take a completely novel approach on encountering a noun phrase (which could be a possible antecedent)



this approach emits a list of permissible pronouns (note that a noun cannot associate with all possible pronouns). Next, when a pronoun is actually encountered, list(s) containing this pronoun is searched. The noun phrase that emitted this list emerges as the right antecedent. If the pronoun does exist in more than one list, a conflict resolution protocol is executed to find out the right antecedent by using a set of rules evaluated with their degree of confidence. We obtain a state-of-the-art performance. In future, we would like to extend our algorithms for co-reference resolution and develop machine learning tools (deep SVM) for morphological analysis and co-reference resolution in resource-scarce languages. Coreference resolution is the task of finding all expressions that refer to the same entity in a text. It is an important step for a lot of higher level NLP tasks that involve natural language understanding such as document summarization, question answering, and information extraction.

Morphological Analysis: Morphological analysis is the process of segmenting words into morphemes and analyzing the word formation. Morphological Analyzers are used in search engines, speech synthesizer, speech recognizer, lemmatization, noun compounding, spell and grammar checker and machine translation. We developed a rule based morphological analyzer for Bangla Text Processing exploiting the principle of finite automata. The automata are used for suffix matching with a predefined list of suffixes collected from the training corpus data. The inflections of numerous Spanish and Bengali words have irregular basis and these are encountered by using a map corresponding to their root words. We also developed a rule-based NER, and it is realized by using untagged data. The rules are formulated with the context information (such as honorific, surname, middle name, relational term, location, and distance) of a named entity. We also conducted the research on the analysis of role of verb, one-expressions, and pro-drop with respect to the Spanish and Bengali languages. These issues are not only important in AR perspective, but also important for several other NLP applications. Since there is no significant computational work on these issues, our work provides a number of important insights in the context of computational linguistics of Spanish and Bengali. A corpus based analysis is provided that investigates quantitative measures of pro-drop and one-expressions in the text. The influence of such issues on the development of the robust anaphora resolution system is considered. Now we are developing a deep learning based morphological analyzer that will be able to learn the rules for identifying the morphemes automatically and segmenting them from the original word. Then, individual morphemes can be further analyzed to identify the grammatical structure of the word.

Industrial Research

Image Processing: I was awarded MITACS Canada industrial internship award three times during my doctoral studies, where I worked on the projects dealing with (i) the early detection/classification of sick birds from thermal images; and (ii) Automatic passenger counting from Light Railway Transit in Edmonton. In the light of different food-related viruses, it is absolutely critical to have a reliable early detection method in the food supply industries/animal farm. My responsibility was to image acquisition (mount web camera at the ceiling and grab the top views to minimize occlusion). I developed a novel variational thresholding method to delineate the warmer regions around the tympanic member of the chicken that is used as a non-invasive biomarker for early detection of Flu. I also implemented parametric curve to automatically detect the passenger heads, followed by optical flow based tracking algorithm and finally validation algorithm to minimize false positive.

Statistical Software Engineering: I completed several industrial projects including: (i) A graph theoretic approach to assess the quality of the class of object oriented software on the basis of a set of metrics and developing a new metric by incorporating the class dependency information; (ii) A bootstrapping method to estimate the efforts required to fix the number of bugs reported by the customer; (iii) A Bayesian approach to evaluate methodology for establishing baselines of build efforts on the basis of SAC (Simple/Average/Complex) methodology; (iv) An integer programming based optimization algorithm for software test case selection; and (v) A regression based software defect prediction methodology.

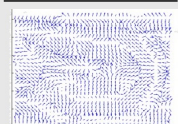
Quality Control and Reliability: Currently I am supervising a group of students in several industrial projects that include: (i) Statistical tools and techniques (pareto analysis, nonparametric control chart and sampling plan) for visual inspection of electronic chip making process; (ii) A multiple sensor based Hidden Markov Model (HMM) for estimating Remaining Useful Life (RUL) of the critical components of a hydraulic system; and (iii) Online removal of chattering by monitoring and detecting natural frequency for a robot guided drill system through kalman filter based control chart.

Future Research Plan

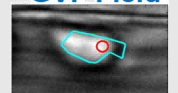
In the long term of my research career, I am interested to solve application-driven large scale (complex) image analysis. In my understanding, solving such complex real-life applications is not a simple task to design smarter and faster algorithms. I believe that alternative theoretical framework could be designed to optimize the solution to these problems.

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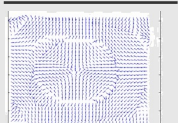
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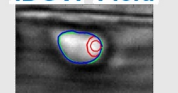
GVF Field



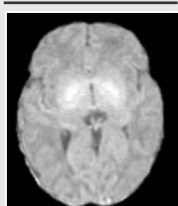
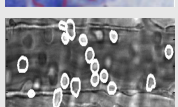
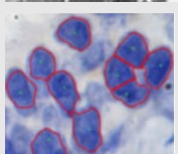
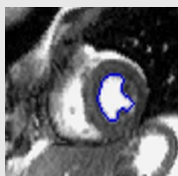
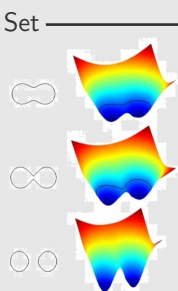
GVF Results



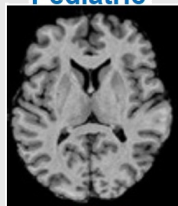
IDGVF Field



IDGVF Results



Pediatric



Adult

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